# A Practical Algorithm for Multiplayer Bandits when Arm Means Vary Among Players

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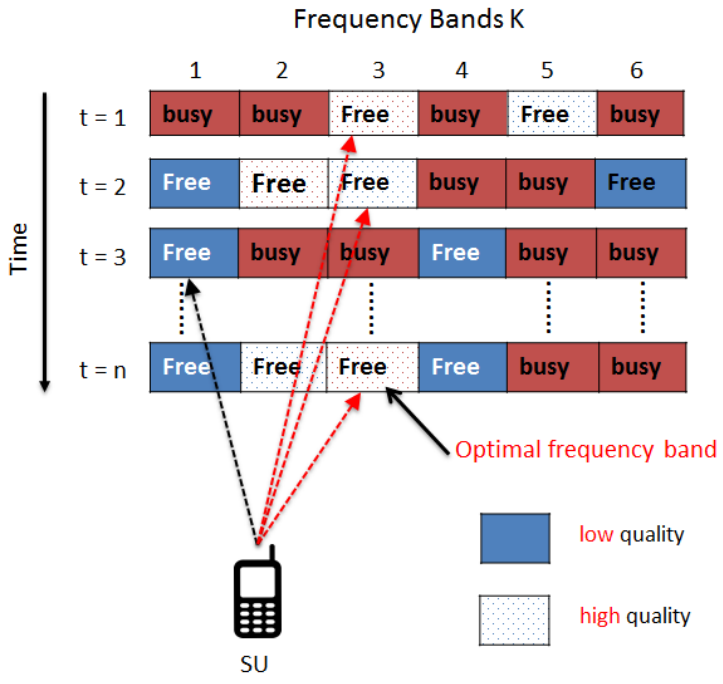
# Motivation: Cognitive Radio

- licensed bands: Opportunistic Spectrum Access

arm $\leftrightarrow$ availability from primary users

- un-licensed bands: IoT communications

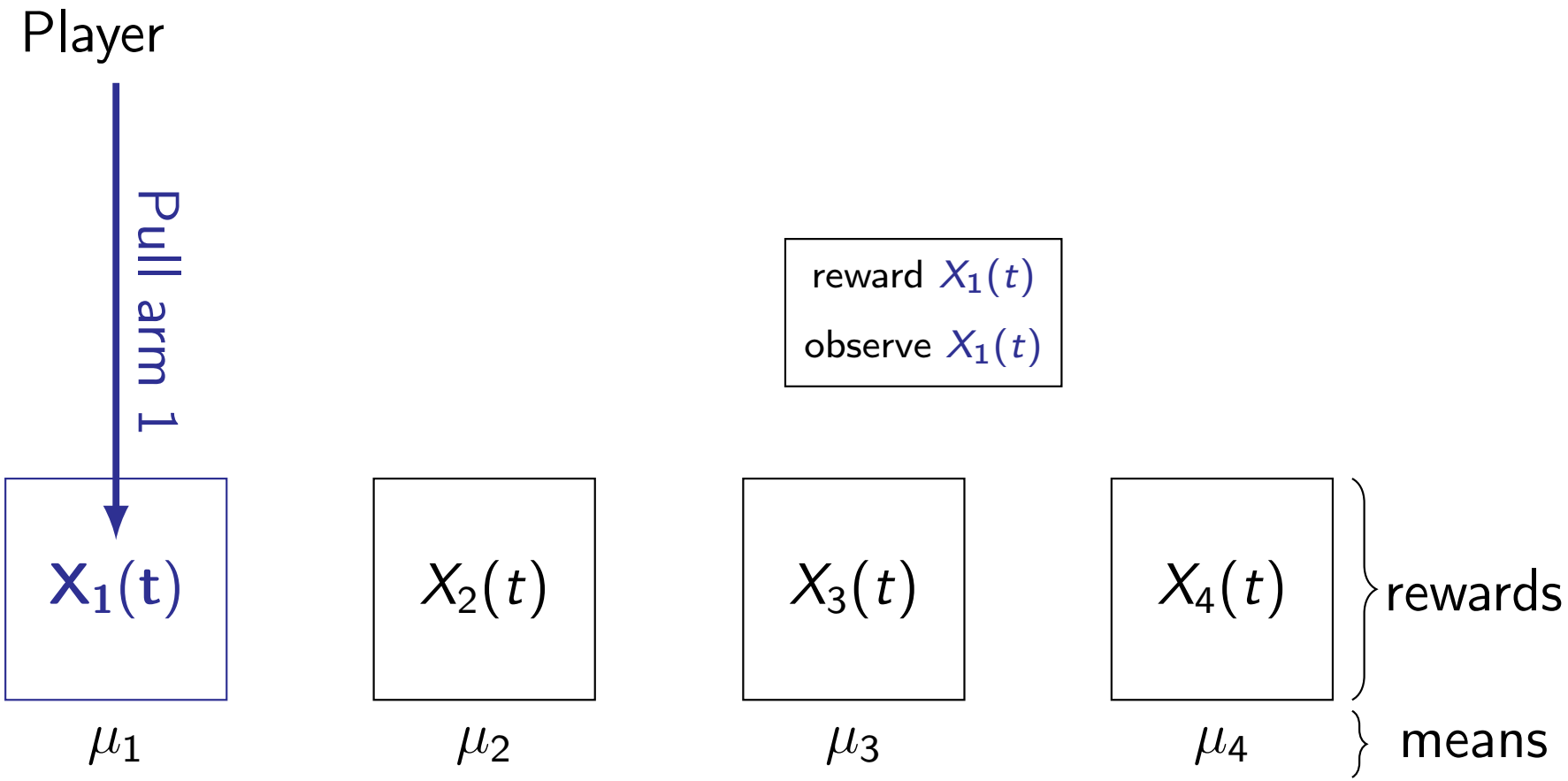
arm $\leftrightarrow$ background traffic




$\rightarrow$ what about **multiple devices**?

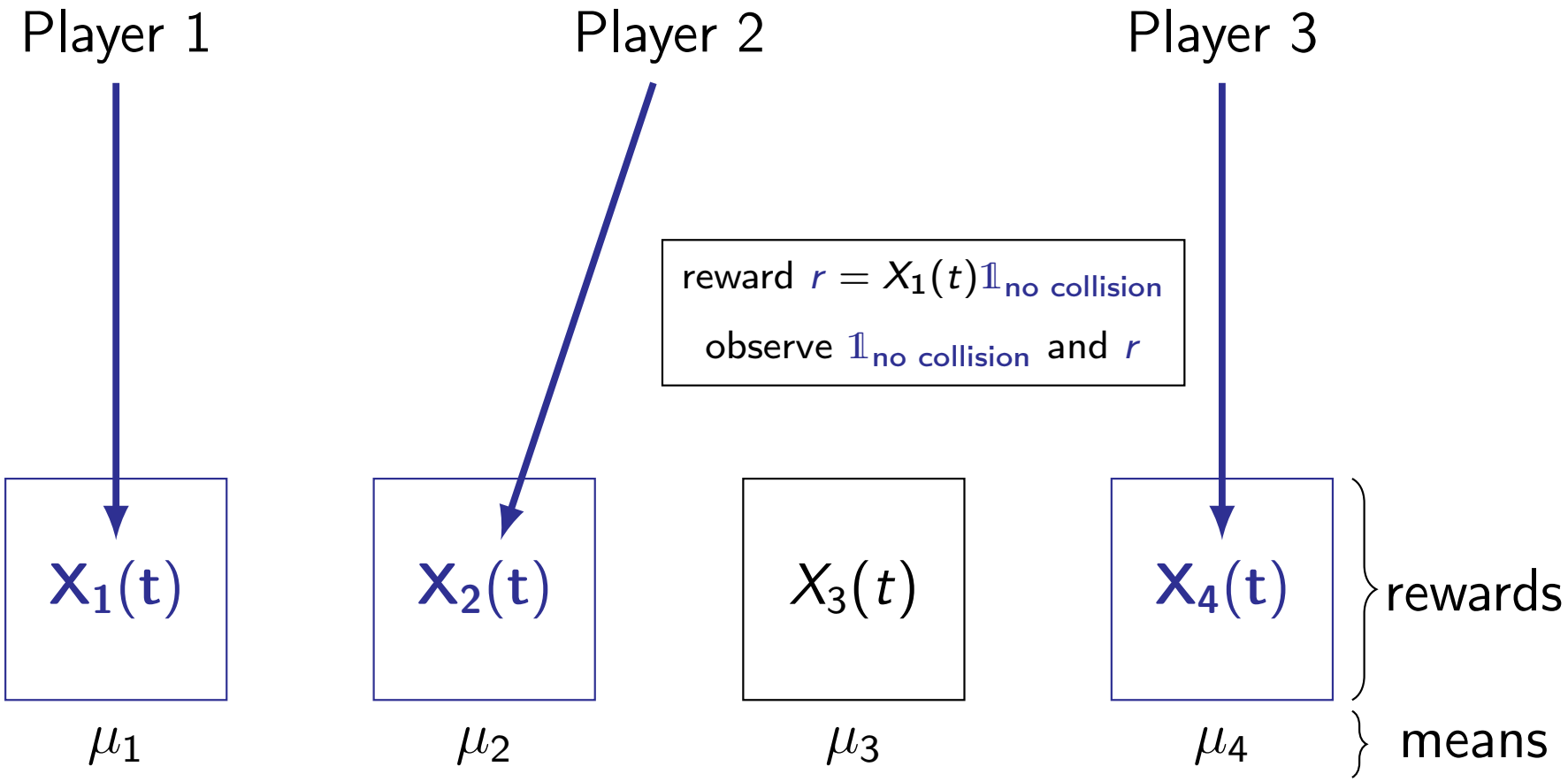
Stochastic bandits

$K$ arms



Stochastic bandits [**Multiplayer**]

$K$ arms, $M$ players



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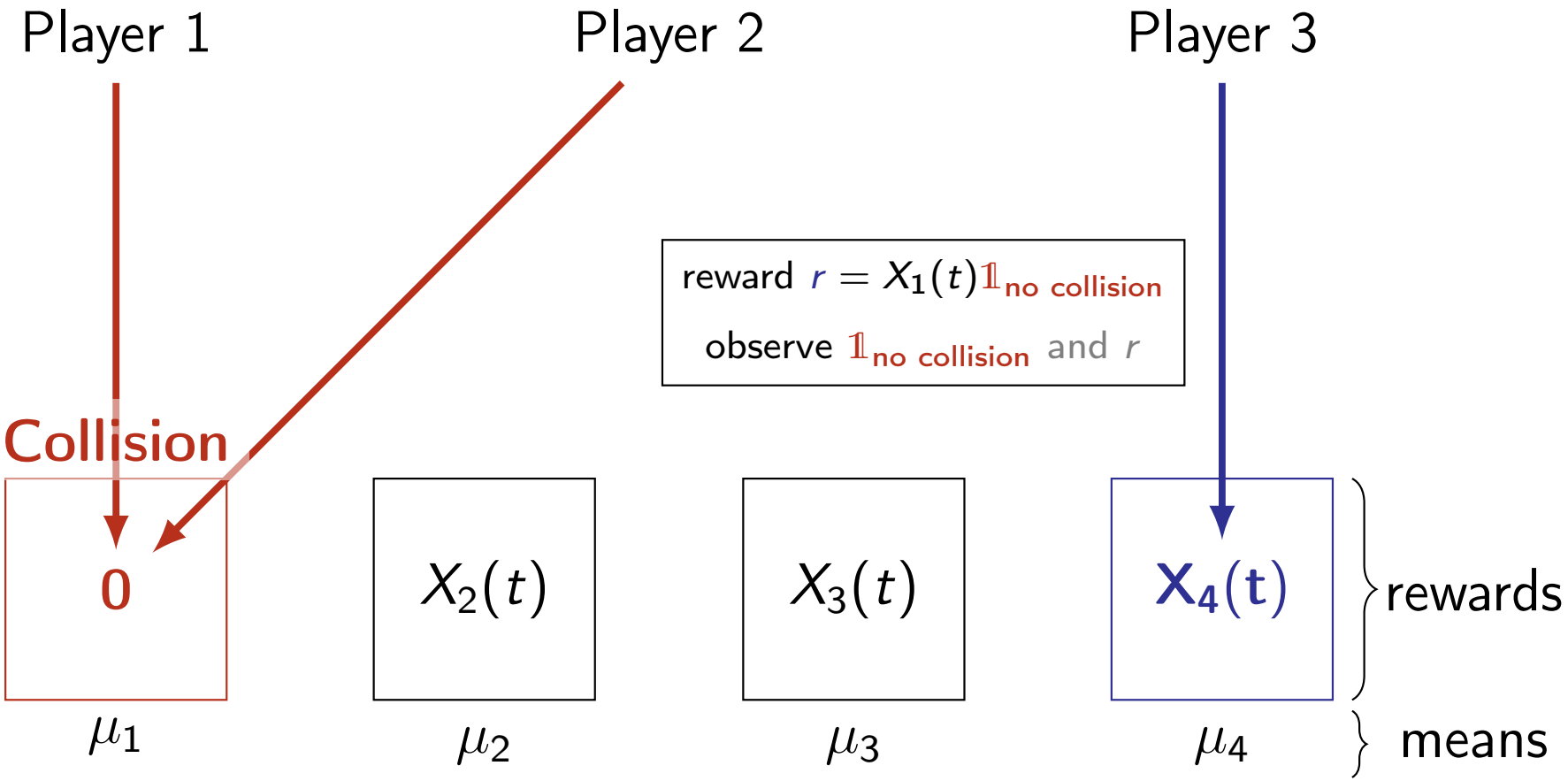

# Model

$M$ players pull at each round arms $A^m(t)$
$K \geq M$ **heterogeneous** arms with reward $X_k^m(t) \in [0,1]$ for player $m$
Arm means $\mu_k^m = \mathbb{E}[X_k^m]$ differ among players

**Goal**: find best player-arm matching
Utility of matching $\pi$: $U(\pi) = \sum_{m=1}^{M} \mu_{\pi(m)}^m$

**Regret:** $R_T = T \max_{\pi} U(\pi) - \mathbb{E}\left[\sum_{t=1}^{T} \sum_{m=1}^{M} \mu_{A^m(t)}^m \mathbb{1}_{\text{no collision on } A^m(t)}\right]$

# Some literature

**Homogeneous** case (same arm means for all players):

- collisions as bit sent between players
- communication based algorithms to reach centralized regret [Boursier and Perchet, 2019, Proutiere and Wang, 2019]

Communication required to determine optimal matching
$\rightarrow$ use communication-based algorithm!

# Some literature

**Heterogeneous** case:

[Bistritz and Leshem, 2018]: algorithm with $\mathcal{O}(\log^{1+\kappa}(T))$ regret, but

- require knowledge of suboptimality gap $\Delta$
- large dependency in problem parameters
- many hyperparameters

[Tibrewal et al., 2019] also proposes communication-based algorithm

- require unique optimal matching
- large dependence in $\Delta$

# Algo presentation

Leader/follower structure to improve communication-based algorithms

**Algorithm 1:** M-ETC-ELIM

**Input:** $T$, $K$, $c$
$m, M \longleftarrow$ Initialization() `// Estimate` $M$ `and ranks for players`
$p \longleftarrow 0$
**repeat**
  Exploration$(p, c)$ `//` $m = 1$ `is leader, other followers`
  Communication$(p, c)$
  $p \longleftarrow p + 1$
**until** *optimal matching* $\pi^*$ *found* or *budget exhausted*
Exploitation: pull $\pi^*(m)$ until $T$ `// optional phase`

# Initialization

- Orthogonalize players: **Musical Chairs** for $K \log(T)$ rounds [Rosenski et al., 2016]
  - Sample arm $k$ uniformly at random
  - $\begin{cases} \text{if collision} \rightarrow \text{continue} \\ \text{otherwise} \rightarrow \text{stick to arm } k \text{ until } K \log(T) \end{cases}$

With proba $1 - M/T$, all players end on different arms

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- Estimate $M$ and attribute rank $m$
  - player on arm $k$ waits for $2k$ rounds
  - *sequential hopping* for $2(K - k)$ rounds
  - number of collisions $= M - 1$
    number of collisions during $2k$ first rounds $= m - 1$

# Communication

Exploration/Communication epochs with length parametrized by $c \in \mathbb{N}$

**Communication trick:** player $m$ sends $(1, 0, \ldots, 1, 1)$ to player $n$ by pulling $(\pi(n), \pi(m), \ldots, \pi(n), \pi(n))$ for some matching $\pi$.

During $p$-th communication phase

- leader $\rightarrow$ follower: list of arms to pull for next exploration
- follower $\rightarrow$ leader: send empirical means in **binary**
  $2^{p^c}$ observations on each arm: $p^c$ bits

$R_T^{\text{comm}} = \mathcal{O}(MKp_T^{c+1})$, where $p_T$ = number of epochs.

# Exploration

During $p$-th exploration: $2^{p^c}$ pulls for each arm sent by leader.

**Player-Arm elimination:**

- Leader computes $\tilde{\pi}^{m,k}$, best matching giving arm $k$ to player $m$ and $\tilde{\pi}^*$, best matching
- For $\tilde{U}(\pi)$ *empirical* utility of $\pi$:

$$\begin{cases} \text{if } \tilde{U}(\tilde{\pi}^*) > \tilde{U}(\tilde{\pi}^{m,k}) + \sqrt{\frac{\log(2KT)}{2^{1+p^c}}}, \text{ eliminate pair } (m,k) \\ \text{otherwise, explore } \tilde{\pi}^{m,k} \text{ next epoch} \end{cases}$$

Do not explore **matchings**, but player-arm pairs
$\rightarrow$ avoid combinatorial regret and complexity

# Regret bound

**Theorem (Regret bound for any $c \in \mathbb{N}^*$)**

$$\mathbb{E}[R_T] = \mathcal{O}\left( MK \left( \frac{M^2 \log(T)}{\Delta} \right)^{1+1/c} \right)$$

Regret $\ln(T)^{1+\delta}$ for any $\delta > 0$

# Regret bound

**Theorem (Regret bound for $c = 1$)**

*a) If optimal matching is **unique**,* $\mathbb{E}[R_T] = \mathcal{O}\left(\frac{M^3 K \log(T)}{\Delta}\right)$

*b)* $\mathbb{E}[R_T] = \mathcal{O}\left(M^{\frac{3}{2}}\sqrt{K \log(T) T}\right)$ *(minimax)*

$\rightarrow$ in practice, take $c = 1$

**Proof of a).**

- unique optim. matching $\implies p_T = \mathcal{O}\big(\log(\log T)\big)$ epochs
- negligible communication regret

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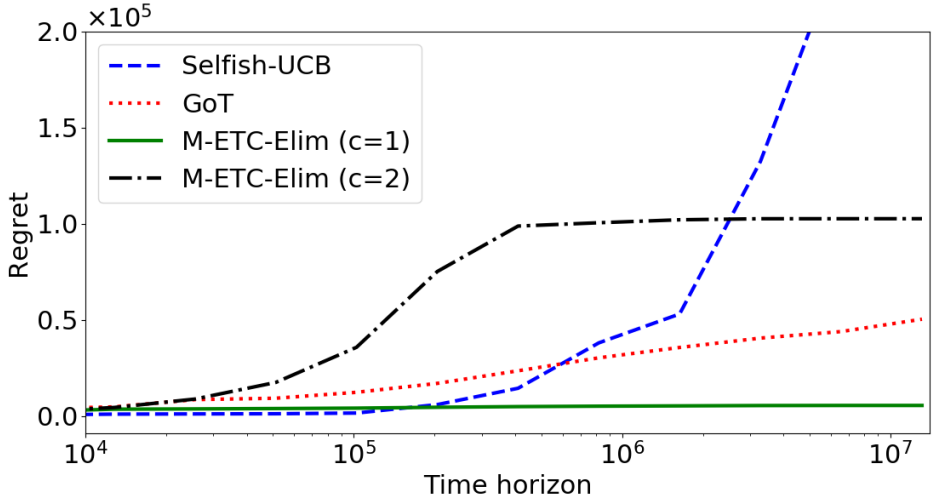
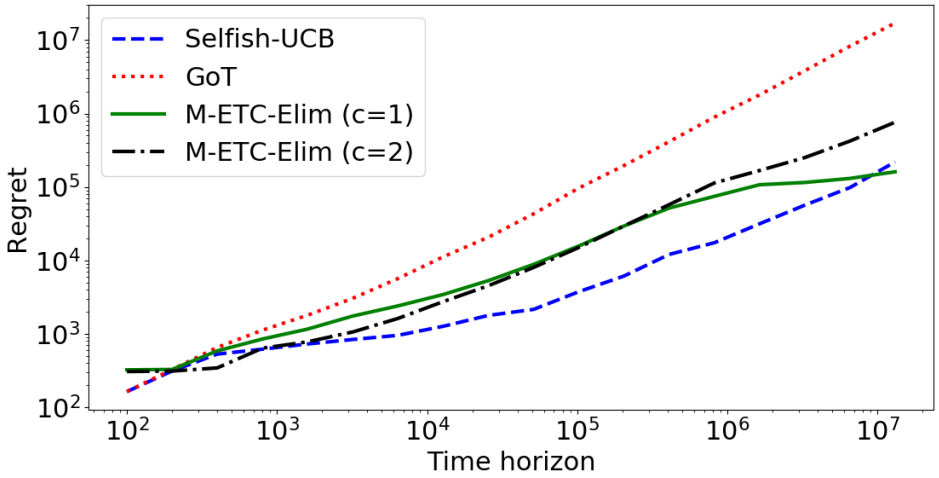
# Experiments




(a) Instance with unique optimal matching ($M = K = 3$)




(b) Harder instance with many optimal matchings ($M = K = 5$)

empirically $\rightarrow$ M-ETC-ELIM with $c = 1$

# Recap

New algo for heterogeneous multiplayer bandits:

- regret $\log(T)^{1+\delta}$ and $\log(T)$ with unique optimal matching
- minimax bound
- require no knowledge on problem parameters
- good empirical performance

**Open question:** heterogeneous setting with no collision observation?

**Thank you!**

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